

Metadata for

Timing matters: Viticultural land use determines responses in structure and function of fungal stream communities across one growing season

Verena C. Schreiner^{1,2, *}, Moritz Link³, Gesa Amelung³, Katharina Ohler³, Romana Salis^{1,4}, Florian Leese¹ & Ralf B. Schäfer^{1,2}

¹ Faculty of Biology, University of Duisburg-Essen, Universitätsstrasse 2, 45141 Essen, Germany

² Research Center One Health Ruhr, University Alliance Ruhr, Universitätsstrasse 2, 45141 Essen, Germany.

³ iES Landau, Institute for Environmental Sciences, RPTU Kaiserslautern-Landau, 76829 Landau, Germany

⁴ Department of Biology and Environmental Science, Linnaeus University, 39231 Kalmar, Sweden

Present address:

Gesa Amelung, Federal Environmental Agency UBA, 06844 Dessau-Roßlau, Germany

Correspondence: verena.schreiner@uni-due.de

All raw data and the corresponding computer code mentioned below can be found under https://github.com/VCSchr/Timing_matters.

- 0_R_Schreiner_et_al_Timing_matters: provides the R Markdown code to calculate the leaf decomposition and the fungal communities across treatments and time points
- 0_R_Schreiner_et_al_Timing_matters.html: knitted R Markdown of the related code
- 1_leaf_weight: leaf mass before and after deployment
 - o Leaf.bag.number: ID of the single leaf bags (categorical)
 - o time.point: coding of the study time points (categorical)
 - o weight_start: dry leaf mass at deployment [g]
 - o stream: name of the streams (categorical)
 - o treatment: name of the treatment (categorical)
 - o replicate: indication of the replicate number (numerical, dimensionless)
 - o number.days: duration time [days]
 - o dry_weight_rest: dry mass of leftover leaf mass after retrieval [g]
 - o ashes_rest: ash mass of leftover leaf mass after retrieval [mg]
 - o dry_weight_OMD: dry mass of OMD leaf disks [mg]
 - o ashes_OMD: ash mass of OMD leaf disks [mg]
 - o dry_weight_conidia: dry mass of conidia leaf disks [mg]
 - o ashes_conidia: ash mass of conidia leaf disks [mg]
 - o dry_weight_ergosterol: dry mass of ergosterol leaf disks [mg]
 - o disks_OMD: number of leaf disks cut for the OMD endpoint (numerical, dimensionless)
 - o disks_conidia: number of leaf disks cut for the conidia endpoint (numerical, dimensionless)
 - o disks_total: number of cut leaf disks (numerical, dimensionless)
- 2_Transport_controls: information on handling controls for leaf decomposition
 - o leaf_bag_number: ID of the single leaf bags (categorical)
 - o time_point: coding of the study time points (categorical)

- type: coding of the type of leaf bags (categorical)
 - weight_60: dry mass of leaf mass after handling [mg]
 - weight_500: ash mass of leaf mass after handling [mg]
 - weight_start_g: initial leaf mass before handling [g]
- 3_mean_temp_treatments: stream temperature for different treatments and time points
 - stream: name of the streams (categorical)
 - sampling: coding of the study time points (categorical)
 - down: mean stream temperature of the downstream / viticultural study locations [°C]
 - up: mean stream temperature of the upstream / forest study locations [°C]
 - move: mean temperature the transplant treatment bags were exposed to [°C]
- 4_conidia_data: data on morphologically identified hyphomycetes
 - stream: name of the streams (categorical)
 - treatment: name of the treatment (categorical)
 - time.point: coding of the study time points (categorical)
 - squares_total: number of total squares on the filter (numerical, dimensionless)
 - squares_examined: fields of the filter which were examined (numerical, dimensionless)
 - filtered_volume: volume of conidia sample filtered [mL]
 - OTHER COLUMNS: number of conidia counted of the taxon named in the respective column head (numerical, dimensionless)
- 5_ITS_DataProp: data on fungi identified via metabarcoding
 - ID: ID of the single metabarcoding samples (categorical)
 - Stream: name of the streams (categorical)
 - treatment: name of the treatment (categorical)
 - type: coding of the type of leaf bags (categorical)
 - timepoint: coding of the study time points (categorical)
 - Miseq.run: code of metabarcoding run (categorical)
 - OTHER COLUMNS: fraction of reads of the ASV named in the respective column head (numerical)
- 6_ITS_Taxonomy_AH: taxonomic information in fungi identified via metabarcoding with assignment for hyphomycetes
 - “ “: ID of the respective ASV (categorical)
 - Kingdom: Kingdom of the respective ASV (categorical)
 - Phylum: Phylum of the respective ASV (categorical)
 - Class: Class of the respective ASV (categorical)
 - Order: Order of the respective ASV (categorical)
 - Family: Family of the respective ASV (categorical)
 - Genus: Genus of the respective ASV (categorical)
 - Species: Species of the respective ASV (categorical)
 - AQH: Coding if ASV is an aquatic hyphomycete (categorical, Y/)
- 7_DADA2_analysis_ITS: R script for DADA2 pipeline, metabarcoding rawdata available upon request
- 8_sample_data: read in sample names for metabarcoding data
 - x: full ID name of the single metabarcoding samples (categorical)
 - ID: ID of the single metabarcoding samples (categorical)
 - Stream: name of the streams (categorical)
 - treatment: name of the treatment (categorical)
 - type: coding of the study time points (categorical)
 - timepoint: coding of the study time points (categorical)
 - Extraction_ID: ID of the DNA extraction (categorical)
 - Forward index: index of the forward primer (categorical)
 - i5 seq: forward primer (categorical)
 - Reverse index: index of the reverse primer (categorical)
 - i7 seq: reverse primer (categorical)

- Miseq run: code of metabarcoding run (categorical)
- 9_ITS_Taxonomy: taxonomic information in fungi identified via metabarcoding with
 - “ “: ID of the respective ASV (categorical)
 - Kingdom: Kingdom of the respective ASV (categorical)
 - Phylum: Phylum of the respective ASV (categorical)
 - Class: Class of the respective ASV (categorical)
 - Order: Order of the respective ASV (categorical)
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